Dimitris Vamvourellis

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Work Experience

Millennium Management — New York, NY
AI Researcher, Equities — *Jul 2025 – Present*

- Research and develop NLP signals for systematic equity strategies from diverse unstructured data, including proprietary internal sources, company disclosures and sell-side research, assessing performance through backtesting and live evaluation.
- Build AI systems that emulate fundamental and event-driven equity research at scale, translating investment theses into systematic trading signals.
- Develop a self-evolving quantitative research agent that autonomously proposes and backtests investment hypotheses in a closed feedback loop, capturing each iteration's evaluation feedback in persistent memory that guides subsequent signal exploration.
- Build AI systems for thematic long/short basket construction and company-specific risk discovery.

BlackRock — New York, NY
Senior AI Researcher, Investment AI Group — *Mar 2021 – Jul 2025*

- Developed sentiment signals using fine-tuned BERT models and deployed them in live systematic equity strategies and trade execution.
- Developed and deployed an ML model for real-time pricing of European and US corporate bonds; reduced OOS MAE by 24% versus the prior production model.
- Developed a multi-agent LLM framework combining financial reasoning and code generation to support investment research, portfolio management and trading.

Intelligencia AI — Cambridge, MA
Data Scientist — *Mar 2020 – Mar 2021*

- Built NLP and ML models to identify emerging areas of innovation in clinical research.

BlackRock — Edinburgh, UK
Analyst, Aladdin Business — *Jul 2017 – Jun 2019*

- Advised portfolio managers on Aladdin's risk modeling, security analytics and portfolio management workflows.

Education

Harvard University — Cambridge, MA
MS in Data Science, GPA: 4.0 — *Aug 2019 – Mar 2021*

The University of Edinburgh — Edinburgh, UK
BS in Computer Science and Mathematics — *Sep 2013 – May 2017*
First Class Honors (Ranked 1st)

Publications

Reasoning or Overthinking: Evaluating Large Language Models on Financial Sentiment Analysis — *Oct 2025*
D. Vamvourellis, D. Mehta
6th ACM International Conference on AI in Finance.

- Compared reasoning and non-reasoning LLMs on zero-shot financial sentiment analysis. Investigated how reasoning depth and prompting strategies affect accuracy and alignment with human judgments.

Company Similarity using Large Language Models — *Oct 2024*
D. Vamvourellis et al.
Symposium on Computational Intelligence for Financial Engineering and Economics (CIFEr).

- Evaluated small and large language models for generating financially meaningful company embeddings from business descriptions in SEC filings.

Enhanced Local Explainability and Trust Scores with Random Forest Proximities — *Oct 2024*
J. Rosaler, D. Desai, B. Sarmah, **D. Vamvourellis** et al.
5th ACM International Conference on AI in Finance.

- Developed a novel approach to explain predictions and quantify confidence of random forest models.

Learning Mutual Fund Categorization using Natural Language Processing — *Oct 2022*
D. Vamvourellis et al.
3rd ACM International Conference on AI in Finance.

- Developed BERT-based mutual fund embeddings from textual descriptions and explored their business applications in fund similarity learning.

BaCOUn: Bayesian Classifiers with Out-of-Distribution Uncertainty — *Jul 2020*
T. Guénais*, **D. Vamvourellis*** et al. *(equal contribution)*
ICML Workshop on Uncertainty and Robustness in Deep Learning.

- Developed a novel Bayesian framework for estimating and decomposing the predictive, in-distribution and out-of-distribution uncertainty of Bayesian neural networks for classification tasks.

Skills

Programming: Python (pandas, PySpark, PyTorch, scikit-learn, PyMC3, Gensim, LangChain, CatBoost), SQL.